7 A proton travelling in a vacuum at a speed of 2.0×10^5 m s^{-1} enters a region of uniform magnetic field of magnetic flux density 3.5 mT . The magnetic field is directed into the plane of the paper.

Initially, the proton is travelling in a direction normal to the magnetic field. The path of the proton while in the magnetic field is the arc of a quarter circle as shown in Fig. 7.1.

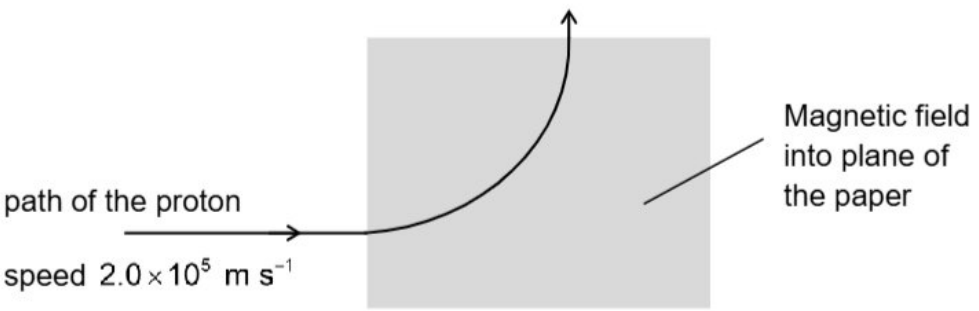


Fig. 7.1

(a) Explain why the path in the magnetic field is an arc of a circle.

.....

.....

.....

.....

..... [2]

(b) (i) Show that the radius of the circular arc is 0.60 m .

[1]

(ii) By considering the change in momentum of the proton, calculate the magnitude of the average force exerted on the proton while it was in the uniform magnetic field.

average force = ... N [4]

[Total: 7]